Touhidul Alam Seyam

MACHINE LEARNING ENGINEER | APPLIED AI RESEARCHER

Chattogram, Bangladesh | +880 1311-104804 | seyamalam41@gmail.com | seyamalam.vercel.app | github.com/Seyamalam | orcid.org/0009-0007-7512-1893

RESEARCH PROFILE

Applied machine-learning researcher and software engineer working across computer vision, tabular modeling, healthcare analytics, agricultural AI, LLM inference, and high-performance computing. Author or co-author of ten source-backed research works, including two IEEE COMPAS papers, two SpringerOpen journal articles, and a 2026 Springer conference chapter. Builds reproducible Python pipelines and production-facing web systems that translate experiments into usable tools.

RESEARCH AND ENGINEERING SKILLS

Machine Learning: scikit-learn, TensorFlow, PyTorch, XGBoost, LightGBM, CNNs, transfer learning, SVM, clustering, feature engineering, cross-validation, model evaluation

Engineering: Python, TypeScript, Next.js, React, Node.js, FastAPI, Django, PostgreSQL, MongoDB, Redis, Docker, Git, Zig, Bun

Research Practice: reproducible experiments, fold-safe validation, benchmark design, data pipelines, academic writing, literature review, publication preparation

RESEARCH EXPERIENCE

Research Assistant | BGC Trust University Bangladesh | Chattogram, Bangladesh | Jul 2023–2025

- Conduct machine-learning experiments and data analysis; implement models; prepare manuscripts, figures, and technical documentation with faculty collaborators.
- Co-authored research across agricultural diagnostics, pulmonary-tuberculosis risk, cardiovascular-risk prediction, malware classification, LLM inference, and Mojo-accelerated clustering.
- Help mentor junior students in research methods and support technical workshops and academic project execution.

Software Engineer | Hello World Communications Ltd | Chattogram, Bangladesh | Aug 2024–Present

- Develop and deploy scalable applications and AI-enabled features using TypeScript, React, Next.js, Node.js, Python, and PostgreSQL.
- Bridge experimental software and production application requirements through code review, system design, deployment, and support.

SELECTED RESEARCH PROJECTS

Smartphone Addiction Prediction | Python, LightGBM, XGBoost, CatBoost | github.com/Seyamalam/playground-series-s6e8

- Built leakage-safe, five-fold out-of-fold pipelines with nested target encoding, matched-fold experiment screening, schema checks, and rank blending.

Kaggriculture Autonomous Agent | Python, Simulation, Evaluation | github.com/Seyamalam/Kaggriculture

- Engineered a deterministic farming agent and reproducible evaluation harness with seat-swapped tournaments, frozen replay corpora, manifest hashing, and regression gates.

bun-scikit | TypeScript, Bun, Zig | github.com/Seyamalam/bun-scikit

- Implemented a scikit-learn-inspired API with native Zig acceleration, parity contracts, benchmarks, documentation coverage checks, and broad model/preprocessing/evaluation support.

PUBLICATIONS

- *"Architectures for AI-Driven Visual Assistance: Evaluating Server-Mediated Mobile and Direct Access Desktop Client Implementations."* Lecture Notes in Networks and Systems, 2026. DOI: 10.1007/978-3-032-15764-5_45
- *"Enhancing Agricultural Diagnostics: Advanced Training of Pre-Trained CNN Models for Paddy Leaf Disease Detection."* Machine Learning Research, 2025. DOI: 10.11648/j.mlr.20251001.11
- *"Next-Generation K-Means Clustering: Mojo-Driven Performance for Big Data."* International Journal of Intelligent Information Systems, 2025. DOI: 10.11648/j.ijiis.20251401.12
- *"Efficient Malware Classification Using Multiprocessing and Bag-of-Words Vectorization."* Advances in Networks, 2025. DOI: 10.11648/j.net.20251201.12
- *"Fine-tuning LLaMA 2 interference: a comparative study of language implementations for optimal efficiency."* arXiv, 2025. DOI: 10.48550/arXiv.2502.01651
- *"AgriScan: Next.js powered cross-platform solution for automated plant disease diagnosis and crop health management."* Journal of Electrical Systems and Information Technology, 2024. DOI: 10.1186/s43067-024-00169-7
- *"Application of Machine Learning K-Means Clustering and Linear Regression in Determining the Risk Level of Pulmonary Tuberculosis."* IEEE COMPAS, 2024. DOI: 10.1109/COMPAS60761.2024.10796963
- *"Enhancing Cardiovascular Risk Prediction Using Support Vector Machines and Advanced Machine Learning Algorithms."* IEEE COMPAS, 2024. DOI: 10.1109/COMPAS60761.2024.10796805
- *"Comparing pre-trained models for efficient leaf disease detection: a study on custom CNN."* Journal of Electrical Systems and Information Technology, 2024. DOI: 10.1186/s43067-024-00137-1
- *"Comparative Performance Evaluation of Classical Machine Learning and Quantum SVM for Heart Disease Prediction using a Quantum-Featured Dataset."* Sonargaon University Journal, 2025. Link: su.edu.bd/web_assets/journal/journal_five/journal3.pdf

EDUCATION

B.Sc. (Hons.) in Computer Science and Engineering | BGC Trust University Bangladesh | Chattogram, Bangladesh | Expected Dec 2026

Research interests: applied machine learning, computer vision, agricultural AI, healthcare analytics, LLM systems, and high-performance ML.

RESEARCH PROFILES

ORCID: 0009-0007-7512-1893 | Google Scholar: scholar.google.com/citations?user=gRkTVYEAAAAJ | GitHub: github.com/Seyamalam